

Incidence Geometry with Polarity and Tiled Surfaces

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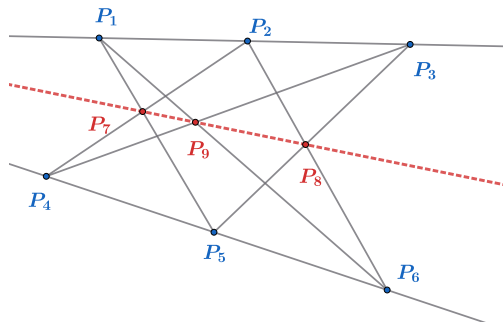
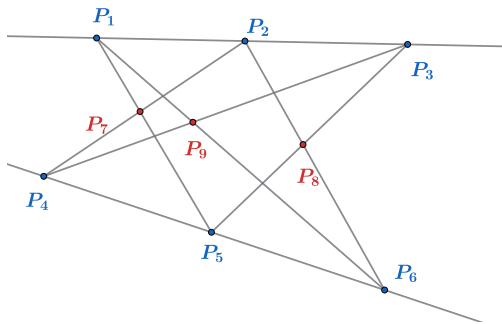
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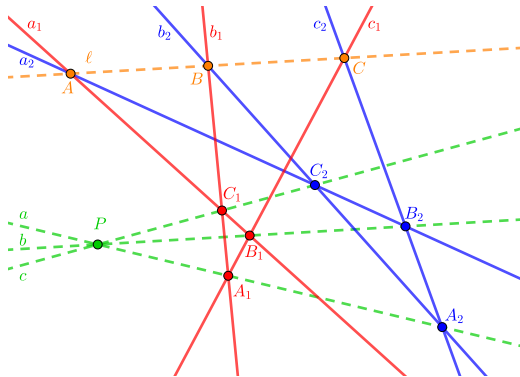
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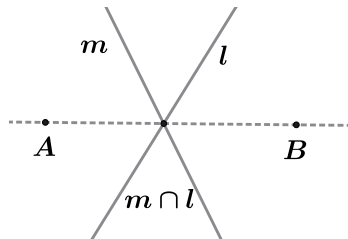
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Incidence Theorem: Pappus's Theorem



Incidence Theorem: Desargues's Theorem





$A, B, m \cap l$: collinear



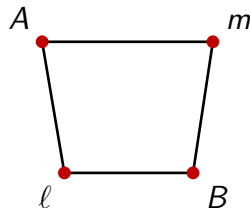
$m, l, (AB)$: concurrent



$$\frac{\langle m, A \rangle \langle l, B \rangle}{\langle m, B \rangle \langle l, A \rangle} = 1$$

This relation can be encoded by
a quadrilateral containing the
four elements A, B, m, l and
Mixed Cross Ratio:

$$\chi(A, B; m, l) = \frac{\langle m, A \rangle \langle l, B \rangle}{\langle m, B \rangle \langle l, A \rangle}$$



coherent tile



$$\chi(A, B; m, l) = 1$$

Master Theorem [Fomin–Pylyavskyy, 2023]

Consider a quadrilateral tiling T of a **closed oriented surface**, whose vertices are colored black and white in bipartite fashion.

black vertex



$$A \in \mathbb{P}$$

associated with a
projective point

For every edge



the point A does not lie
on the hyperplane h :

$$A \notin h$$

white vertex



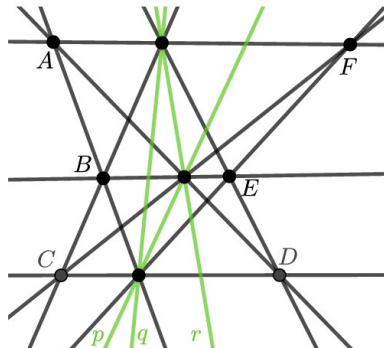
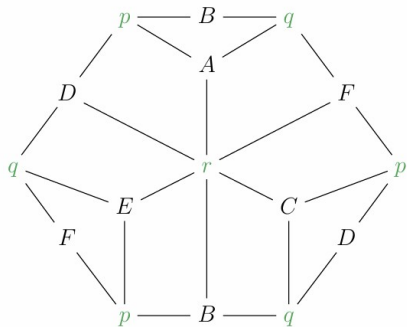
$$h \in \mathbb{P}^*$$

associated with a
hyperplane

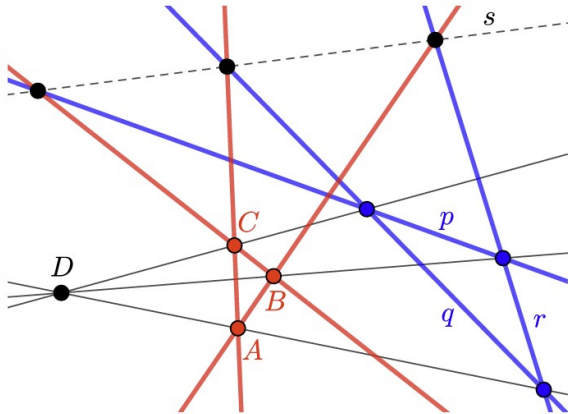
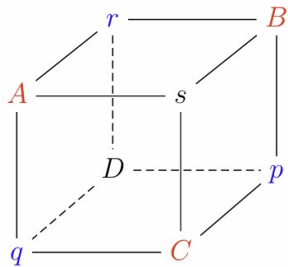
Master Theorem

If all tiles of T , with the exception of one, are coherent, then the remaining tile is coherent as well.

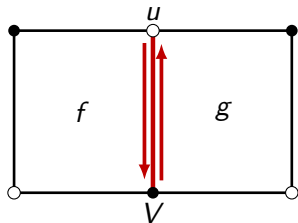
Master Proof of Pappus's Theorem



Master Proof of Desargues's Theorem



Proof of the Master Theorem



$$f : u \rightarrow V, \quad g : V \rightarrow u$$

same edge uV , opposite orientations

For the shared edge uV ,

$$\langle u, V \rangle \langle u, V \rangle^{-1} = 1.$$

$$\prod_{\text{faces } f} \chi(f) = 1$$

$$\chi(f) = 1 \iff f \text{ coherent.}$$

$$\chi(f) = 1 \ (f \neq f_0) \implies \chi(f_0) = 1$$

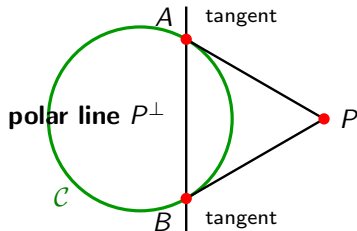
Attention:

There is a 2-coloring of the Surface
and the surface must be **oriented**. (Question: How
about nonoriented case?)

Projective Polarity

$$\beta(\mathbf{x}, \mathbf{y}) = x_1y_1 + x_2y_2 - x_3y_3, \quad \mathcal{C} : \beta(\mathbf{x}, \mathbf{x}) = 0$$

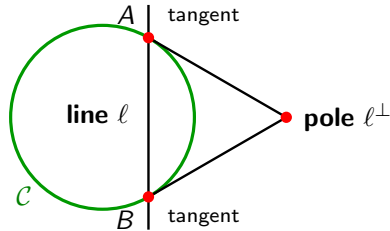
Point $\longrightarrow$ Polar Line



$$P \longrightarrow A, B \longrightarrow P^\perp = AB$$

$$\beta(A, P) = \beta(B, P) = 0$$

Line $\longrightarrow$ Pole

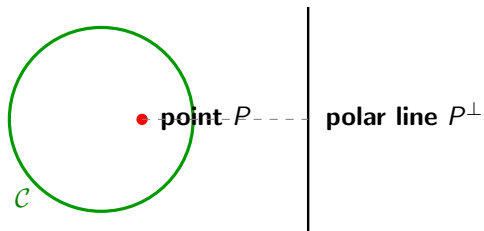


$$\ell \cap \mathcal{C} = \{A, B\} \longrightarrow \ell^\perp = PA \cap PB$$

$$\beta(A, P) = \beta(B, P) = 0$$

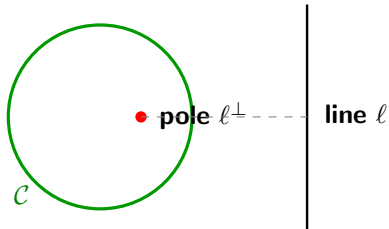
Projective Polarity

Interior Point $\longrightarrow$ Exterior Polar Line



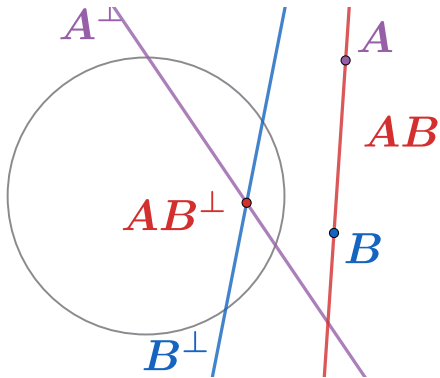
$$P \in \text{int}(\mathcal{C}) \implies \boxed{P^\perp \cap \mathcal{C} = \emptyset}$$

Exterior Line $\longrightarrow$ Interior Pole



$$\ell \cap \mathcal{C} = \emptyset \implies \boxed{\ell^\perp \in \text{int}(\mathcal{C})}$$

Projective Polarity

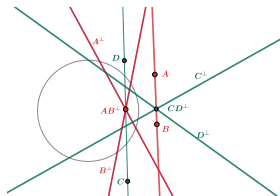
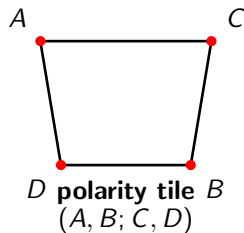


$$A, B \in (AB)$$



$$(AB)^\perp = A^\perp \cap B^\perp$$

Coherent Polarity Tile and Mixed Cross Ratio of Polarity



$A, B, (CD)^\perp$ are collinear.

$$\text{MCRP: } \chi_\beta(A, B; C, D) = \frac{\beta(A, C)\beta(B, D)}{\beta(B, C)\beta(A, D)}$$

$$\chi_\beta(A, B; C, D) = 1$$

coherent



$$AB \perp_\beta CD$$



$A, B, (CD)^\perp$ are collinear



$C, D, (AB)^\perp$ are collinear



$A^\perp, B^\perp, CD$ are concurrent

The Master Theorem of Polarity [Jiang-L.-L.26+]

Polarity Master Theorem.

Consider a labeled quadrilateral tiling Q of a connected closed surface (**Oriented or Nonoriented**).

Assume

vertices labeled by projective points, adjacent labels not β -conjugate,

closed cycle γ : $\begin{cases} \text{same orientation on return} & \implies |\gamma| \text{ even,} \\ \text{opposite orientation on return} & \implies |\gamma| \text{ odd.} \end{cases}$

Then

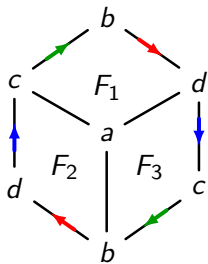
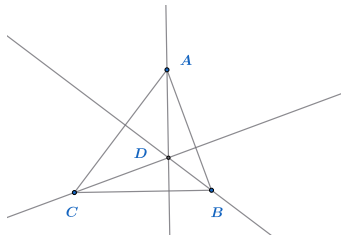
$$\prod_{\text{faces } f} \chi_{\beta}(f) = 1$$

Consequently

$$f \text{ coherent} \iff \chi_{\beta}(f) = 1,$$

$$\text{all but one face coherent} \implies \text{remaining face coherent}$$

Example 1: Hesse Theorem and Its Master Proof



Hesse tiling on $\mathbb{RP}^2$

$$F_1 : ab \perp_\beta cd,$$

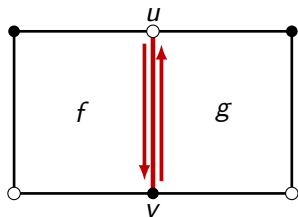
$$F_2 : ad \perp_\beta bc,$$

$$F_3 : ac \perp_\beta bd.$$

Hesse Theorem

$$\left. \begin{array}{l} F_1 \text{ coherent} \\ F_3 \text{ coherent} \end{array} \right\} \implies F_2 \text{ coherent.}$$

Proof of the Master Theorem of Polarity



$f : u \rightarrow v, \quad g : v \rightarrow u$

same edge uv , opposite orientations

Oriented Surface:
Same as Fomin, Pylyavskyy's

Nonoriented Surface:

$S \xrightarrow{\text{Orientation double cover}} \tilde{S}$

do cancellation to $\tilde{S}$

Coherent Polygons [Fomin–Pylyavskyy, 2023]

Consider a topological disk $\mathbf{P}$ with $2n$ marked points on its boundary, labeled cyclically by

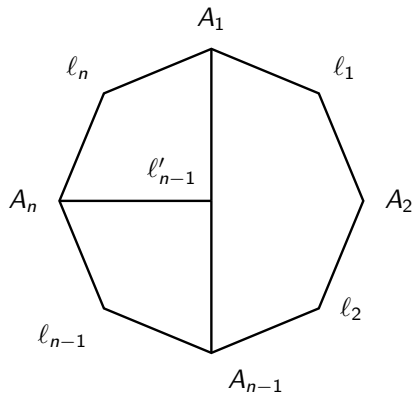
$$A_1, \ell_1, A_2, \ell_2, \dots, A_n, \ell_n.$$

We call $\mathbf{P}$ **coherent** if

$$(A_1, \dots, A_n; \ell_1, \dots, \ell_n) = 1,$$

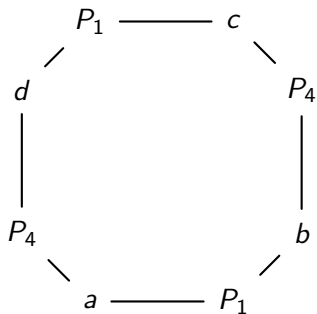
where

$$(A_1, \dots, A_n; \ell_1, \dots, \ell_n) = \frac{\langle \mathbf{A}_1, \ell_1 \rangle \langle \mathbf{A}_2, \ell_2 \rangle \cdots \langle \mathbf{A}_n, \ell_n \rangle}{\langle \mathbf{A}_2, \ell_1 \rangle \langle \mathbf{A}_3, \ell_2 \rangle \cdots \langle \mathbf{A}_1, \ell_n \rangle}.$$

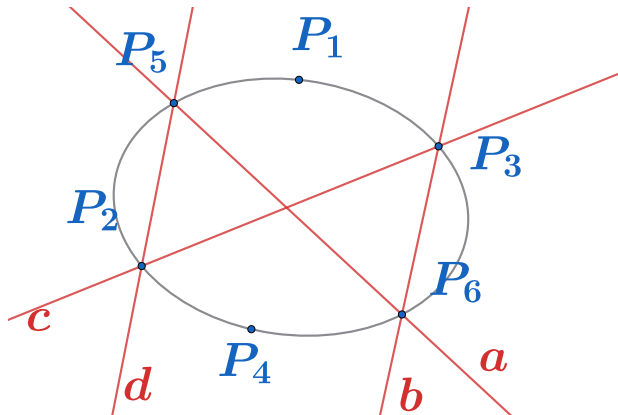


alternating labels A_i, ℓ_i around the boundary

Example 2: Coherent Polygons of a Conic



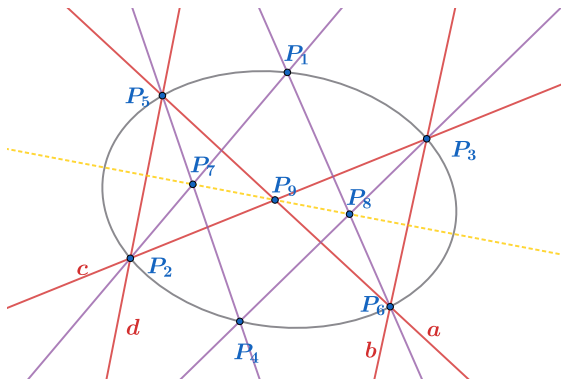
A coherent $2n$ -gon, $n = 4$.



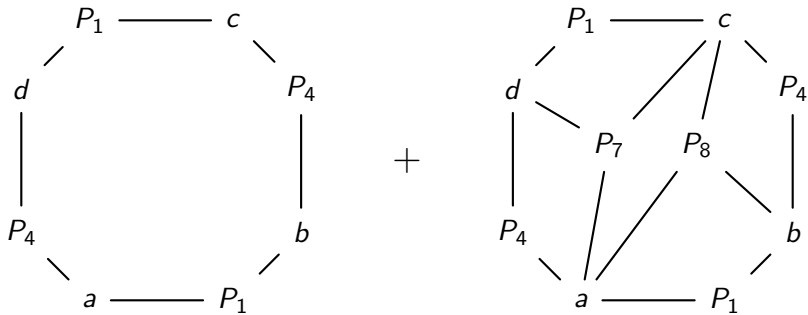
Example 3: Pascal Theorem

Pascal Theorem. Let $P_1, P_2, \dots, P_6$ be six points on a conic. Then

$$P_7 = P_1P_2 \cap P_4P_5 \quad P_8 = P_2P_3 \cap P_5P_6 \quad P_9 = P_3P_4 \cap P_6P_1 \quad \Rightarrow P_7, P_8, P_9 \text{ collinear.}$$



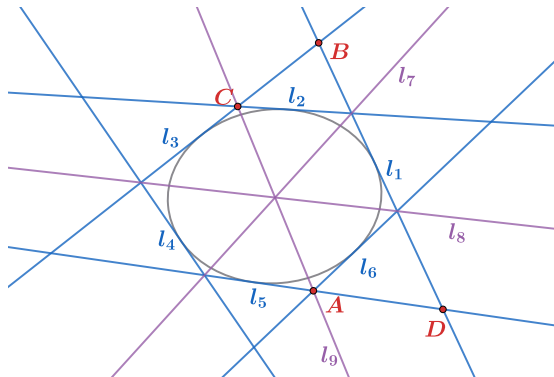
Proof of Pascal Theorem



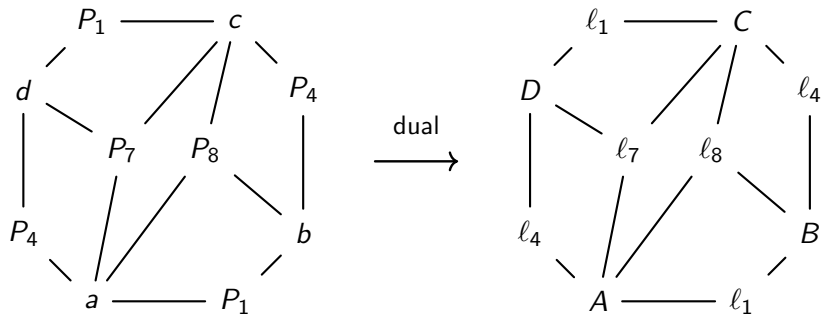
Example 4: Brianchon Theorem

Brianchon Theorem. Let $l_1, l_2, \dots, l_6$ be six tangents on a conic. Then

$$l_7 = (l_1 \cap l_2) \cdot (l_4 \cap l_5) \quad l_8 = (l_2 \cap l_3) \cdot (l_5 \cap l_6) \quad l_9 = (l_3 \cap l_4) \cdot (l_6 \cap l_1) \quad \Rightarrow l_7, l_8, l_9 \text{ concurrent.}$$



Proof of Brianchon Theorem



The twist restores global cancellation

Orientation-twisted

Every even cycle is orientation-preserving.

Every odd cycle is orientation-reversing.

Polarity Master Theorem

$$\prod_{f \in F(Q)} \chi_f(f) = 1.$$

Hence coherence of all but one face forces coherence of the last face.

A Klein Bottle Tiling

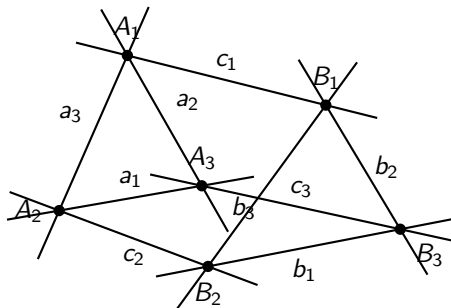
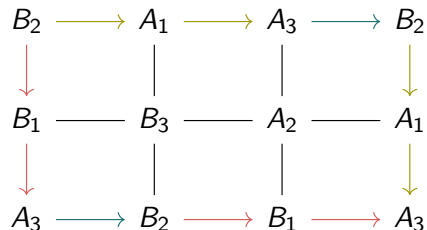
For $A_1, A_2, A_3, B_1, B_2, B_3$, define

$$a_i = A_j A_k, \quad b_i = B_j B_k, \quad c_i = A_i B_i, \quad \{i, j, k\} = \{1, 2, 3\}.$$

The six conditions are

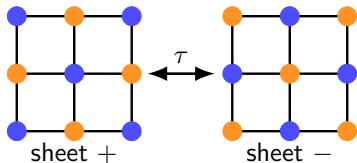
$$a_i \perp_\beta c_i, \quad b_i \perp_\beta c_i \quad (i = 1, 2, 3).$$

Any five of these conditions imply the sixth one.



Orientation Double Cover

Q : Non-orientable tiling, $\tilde{Q}$: orientation double cover, τ : sheet involution



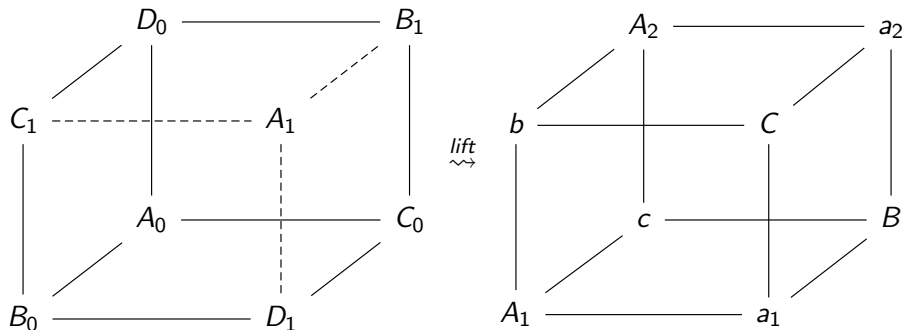
$\forall$ two-coloring on the lifted tiling, τ switches colors.

A polarity incidence theorem lifts to a self-dual incidence theorem

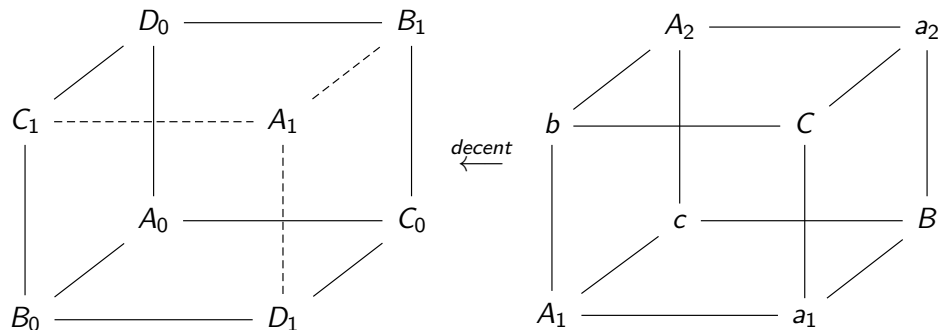
$$\boxed{\bigwedge_{g \in F(Q) \setminus \{f\}} I(g) \implies I(f)} \stackrel{\text{lift}}{\rightsquigarrow} \boxed{\bigwedge_{g \in F(\tilde{Q}) \setminus \{\tilde{f}, \tau\tilde{f}\}} I(g) \implies (I(\tilde{f}) \iff I(\tau\tilde{f}))}$$

Hesse: hemicube on $\mathbb{RP}^2$ $\longleftarrow$ Desargues: cube on S^2

Möbius band $\longleftarrow$ annulus.



Descent is controlled by an admissible involution



- fixed-point-free and orientation-reversing;
- color-swapping: $c \circ \tau = c + 1$;
- such that its orbit cell structure is a regular quadrangulation.

Theorem (Non-commutative Master Theorem, Pylyavskyy–Skopenkov 26, Izosimov 26)

If an incidence theorem has a sphere surface-tiling proof, then it is true over any skew field.

Proposition (Fomin–Pylyavskyy 23)

Pappus' theorem has a torus-tiling proof but is false over any skew field.

Theorem (Jiang–L.–L.26+)

For division ring D , σ -hermitian h , tiling surfaces restrict generalization of D, σ :

- S^2 : arbitrary (D, σ) ;
- Σ_g , $g \geq 1$: D commutative, arbitrary σ ;
- N_k , $k \geq 1$: D commutative and $\sigma = id$.

Hermitian Hesse analogue fails

Over $(\mathbb{C}, -)$, one Hermitian configuration has

$$A = (1, 0, 0),$$

$$B = (1, 1, 0),$$

$$C = (1, i, 1),$$

$$D = (1, i, -1 - i).$$

$AB \perp_h CD, AC \perp_h BD$ but $AD \not\perp_h BC$

Fundamental Group and Euler Genus

Define

$$sl_{G_I}^{odd}(\delta_I) = \min\{k : \delta_I = z_1^2 \cdots z_k^2, z_i \text{ are odd}\},$$

$$cl_{G_I}^{even}(\delta_I) = \min\{g : \delta_I = \prod_{i=1}^g [a_i, b_i], a_i, b_i \text{ are even}\}.$$

Theorem (Euler genus formula, Jiang–L.–L.26+)

$$eg_{el}(I) = tl_{G_I}(\delta_I) = \min\left\{sl_{G_I}^{odd}(\delta_I), 2cl_{G_I}^{even}(\delta_I)\right\}.$$

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